

RESULTS – FRONT END

Team Number
T.N. - 13

Guide Name
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1. Home Page

localhost:5000

Heart Disease Prediction Using Ridge and Lasso Regularized Logistic Regression

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Try the Prediction

Overview | Dataset | EDA | Preprocessing | Algorithms

Overview

The Challenge
Heart disease needs early identification. Real clinical data has missing values, duplicates and outliers, and different feature scales reduce model reliability.

Our Objective
Analyse the dataset through EDA, apply preprocessing (duplicate removal, imputation, scaling) and build Ridge, LASSO and Elastic Net classifiers.

Case-Based Scenario
A 55-year-old patient visits with chest discomfort and shortness of breath. Age, sex, chest pain type, blood pressure, cholesterol, maximum heart rate, exercise-induced angina and more are recorded, preprocessed and passed to the ML model, which predicts whether heart disease is likely.

2. Dataset & EDA

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EDA Findings

Structure
Shape, columns, data types and descriptive statistics.

Quality
Missing values, duplicates, unique values.

Patterns
Distributions, target-wise plots, correlations, boxplots, skewness, kurtosis, IQR outliers.

Correlation with target

Data quality
Missing values in several attributes plus duplicate rows need handling.

Strong signals
cp, thalach, slope (positive); exang, oldpeak, ca, thal (negative).

Balanced target

3. Prediction Page

localhost:5000

Predict Heart Disease

Age 55	**Sex** Male
Chest pain type (cp) 0-3: 0	**Resting BP (trestbps)**: 140
Cholesterol (chol): 240	**Fasting blood sugar >120**: No
Resting ECG 0-2: 1	**Max heart rate (thalach)**: 150
Exercise angina: No	**Oldpeak**: 1.5
Slope 0-2: 1	**Major vessels (ca)** 0-4: 0
Thal 0-3: 2	

Predict

Empty fields are filled with the training median, same as the pipeline. This is a student project, not medical advice.

4. Additional Screens

localhost:5000

Model	Accuracy	Precision	Recall	F1	Specificity
Ridge (L2)	0.8195	0.7698	0.9238	0.8398	0.7100
LASSO (L1)	**0.8293**	**0.7823**	**0.9238**	**0.8472**	**0.7300**
Elastic Net	**0.8293**	**0.7823**	**0.9238**	**0.8472**	**0.7300**

Model Comparison

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Ridge (L2)

	Pred: No	Pred: Yes
Actual: No	71	29
Actual: Yes	8	97

ROC-AUC 0.9317 · 29 false positives, 8 false negatives

LASSO (L1)

	Pred: No	Pred: Yes
Actual: No	73	27
Actual: Yes	8	97

ROC-AUC 0.9346 · 27 false positives, 8 false negatives

Elastic Net

	Pred: No	Pred: Yes
Actual: No	73	27
Actual: Yes	8	97

ROC-AUC 0.9346 · 27 false positives, 8 false negatives

Confusion Matrices

★ Key Front End Features

- ✔ Responsive single-page website
- ✔ Sticky navigation to every section
- ✔ EDA summary with correlation chart
- ✔ Patient form with 13 clinical inputs
- ✔ Predictions from Ridge, LASSO and Elastic Net
- ✔ Model comparison and confusion matrices
- ✔ Flask + scikit-learn backend
- ✔ Automatic light / dark theme

Designed to deliver a clear, user-friendly experience for heart disease risk prediction.